Reproduction of a Research Claim from Rinaldi et al. (2016), from Cognition

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Final Report

December 25, 2020

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Project ID: ("Rinaldi_Cognition_2016_Kj9d - Shaki - Computational Reproduction - 5196")
OSF project: (https://osf.io/8btme/)
Preregistration: (https://osf.io/jhyna/)

Claim Summary

The claim selected for reproduction from Rinaldi et al (2016; experiment 3) is that visuomotor routines of both finger counting (FC) and reading direction (RD) influence the representation of non-numerical ordered sequences (days of the week), so the fastest mappings result when both routines are compatibly oriented. Reaction times were corrected for finger baseline, and those more than two standard deviations from the overall mean were excluded from the analysis. A 2 x 2 x 2 repeated measures ANOVA with days-direction (thumb-little finger, little finger-thumb), hand posture (prone, supine) and language (English, Hebrew) as within-subjects variables was carried out, yielding a significant triple interaction, such that the compatible mappings (Mapping A for English, Mapping G for Hebrew) were the fastest.

Focal inferential test: A significant triple interaction, $F(1, 19) = 16.69$, $p < .002$, partial eta squared = .47, such that the compatible mappings (Mapping A for English, Mapping G for Hebrew) were the fastest.

Reproduction Criteria

The reproduction team has been provided access to the original dataset A successful reproduction of the Rinaldi et al., (2016) results would be a significant triple interaction of the above ANOVA, such that the compatible mappings (Mapping A for English, Mapping G for Hebrew) will demonstrate the fastest performance.

Reproduction Result

A significant triple interaction was found, $F(1, 19) = 6.76$, $p = .018$, partial eta squared = .262. However, in contrast to the original findings for English conditions (fastest mapping A), in the reproduction attempt mappings C and D were fastest than mapping A (although not statistically different in the post-hoc Scheffè comparisons). The reproduced pattern of the Hebrew conditions (fastest mapping G) was similar to the original findings (although not statistically different in the post-hoc Scheffè comparisons).

Thus, this reproduction attempt of the claim was failed according to the SCORE criteria.

Deviations from the Preregistration

There are no known deviations from the original study.

Exploratory findings:

A note from the original author, Luca Rinaldi (provided by SCORE) may explain the failure of reproduction attempt:

"The database uploaded with our Cognition article is the only raw database we saved; when we re-run the analysis (starting from the raw database uploaded in Cognition), we could replicate the main findings for all experimental tasks, but with different results (i.e., numbers in the statistics).

We are, nevertheless, sure that the corrected RTs reported are the right ones (indeed, with this dataset we could fully reproduce the results) and that some errors must have occurred in the handling of the raw database (i.e., possibly when relabeling conditions and Hebrew terms to English) for its publication in Cognition.

Considering the data analysis, however, a piece of information that is not stated in the article and that may be informative for your attempt is that we adopted a 2SD individual cutoff on RTs for both the motor baseline and experimental tasks."

Description of materials provided:

The following materials are publicly available on the OSF site:

- The final report. Filename:
  - SCORE Report - Rinaldi_Cognition_2016_Kj9d_5196 – Shaki
- Two aggregated files:
  - Agg motor All
  - Agg experimental All

- The **data analysis script,** provided as a SPSS file and PDF. This script transforms the raw data into the analysis data set and executes the analyses to get to the final results. Filename:
    - Syntax All. Sps
    - Syntax All.pdf
- The **output** from the SPSS analyses, available as a pdf and .spv (SPSS output) files. Filenames:
    - Focal ANOVA All.spv
    - Focal ANOVA All.htm

Citation

Rinaldi, L., Di Luca, S., Henik, A., & Girell, L. (2016). A helping hand putting in order: Visuomotor routines organize numerical and non-numerical sequences in space. *Cognition*, 152, 40-52.